

BITS Pilani WILP — M.Tech Dissertation Outline Viva

Project Title: GFIS (Green Fuel Intelligence System)

Researcher: Akash Shivadas Chatake (BITS ID: 2024AB05226)

Examiner: Dr. Mohana Murali G. (BITS Pilani)

Co-Examiner / Representative: Dr. Deepti

Location/Platform: BITS Pilani Virtual Viva Bridge

Part 1: Project Proposal & Outline Defense (Session 1)

Dr. Mohana Murali G. (MG): Can you hear me? Hello!

Akash Shivadas Chatake (CS): Yes, sir.

MG: Can you hear me?

CS: Yes, sir. Can you hear me?

MG: Yeah, okay. Yeah, I can hear you. Now it's clear. So please switch on your webcam and—yeah, great. And you can start the presentation.

CS: Good morning, Sir. I'm Akash. Let me introduce myself first. I'm a lecturer in a polytechnic institute in Solapur, and this project is in two phases.

MG: But then... Okay. Just one minute, just one minute, Akash. I'll switch off my fan. I don't know whether it is making noise. Just a second.

CS: Okay.

(Brief pause as MG switches off the fan; presentation slides load on screen)

MG: Okay.

CS: This presentation covers the AI core, soft sensing mathematics, and our validation engine. This is the academic research contribution of this dissertation.

Why "physics-guided"? Because machine learning doesn't understand biology and physics. Left to itself, standard ML will show us that the gasification process continuously increases as we increase pH, increase temperature, and increase the feedstock loading. But practically, that is not possible. There are strict limitations of biology and physical laws in this world.

So, I formulated those constraints into a single custom loss equation. It combines standard losses with biophysical penalty boundaries, allowing the system to predict gasification within the strict rules of physics and biology.

Let us look at the biological black box problem. The process is governed by temperature, pH, OLR (organic loading rate), and Volatile Solids (VS).

MG: Okay.

CS: What was the research gap I was discussing earlier?

Traditional mathematical modeling relies on ADM1 (Anaerobic Digestion Model No. 1). It uses dozens of coupled differential equations. It is mathematically rigorous but computationally too heavy for real-

time edge computing. That's why we are proposing our framework.

On the other hand, pure Machine Learning lacks process sensing. It is a physics-blind modeling approach that learns numerical correlation without understanding conservation laws and biological thermodynamics. This creates an extrapolation danger: if presented with unseen, high OLR inputs, pure ML will blindly predict impossible methane yields.

MG: Yeah.

CS: What is our solution? It is a hybrid architecture combining spatial tabular learning with temporal sequence models. We use the XGBoost library for the tabular modeling, and for our dataset, right now we are using open-source data and soft-sensing simulated data which represents our baseline.

But practically, our institute is backed by a major industrial ecosystem—a cooperative sugar factory in Maharashtra. They have a massive crushing capacity of 15,000 to 20,000 metric tons daily (TCD). They have offered me sponsorship and research support for this project. In fact, we have already filed a copyright application for this framework and are actively proceeding toward a patent once the prototype is fully built.

This sugar factory houses a massive industrial biogas plant. I also have an empirical micro-scale pilot running in my village—a small plant designed for a 5-to-6 person family. I am gathering real-time operational data from both: the micro-scale village plant and the massive industrial sugar factory plant.

The industrial biogas plant at the factory is scheduled to commence its next operational cycle at the end of June. So, I will have access to their historical databases plus live current data using our IoT setups.

This data is used to validate our Soft Sensor, which estimates unmeasurable stability indices—specifically the Volatile Fatty Acids (VFA) to Alkalinity (ALK) ratio—in real-time using easily accessible physical inputs. This feeds into our Digital Twin simulator, which is a software sandbox powering risk-free "What-If" scenario execution on our portal. Let me show you...

MG: Akash, Akash, I think our time is up for this slot. Would 11:45 AM be fine for you to reconvene?

CS: Yeah, yeah. Yes, sir.

MG: We'll continue at 11:45 because I want to understand more about this, and I also have some valuable inputs that I want to discuss and give to you. This is really, really very good. I'm so happy about this. So we'll talk again at 11:45. We can use the exact same bridge, Akash. Okay? Yes.

CS: Okay, okay. No problem at all, sir. Thank you, sir. Thank you, madam.

MG: Yeah, thanks, Dr. Deepti, for joining. Yeah.

Part 2: Deep-Dive Discussion & Academic Breakthrough (Session 2)

(Reconnected on the same virtual bridge at 11:45 AM)

MG: Akash, we can continue.

MG: Dr. Deepti, can you hear me?

MG: Akash, you are on mute.

CS: Hello, sir! Am I audible?

MG: Yeah, you are audible, Akash.

CS: Okay, great. I will start the presentation window. Is the presentation visible, sir?

MG: Oh yeah, it is visible, yes.

CS: Okay, excellent.

(Akash completes the presentation walkthrough of the math, soft sensor states, digital twin sandbox, and validation metrics)

MG: Excellent. You've brought the real chemistry and anaerobic biology together in a true hybrid machine learning approach, right? See if some other model variations can be tested too. Maybe. I don't know, just think about it as you build. But yes, this is the way.

CS: Yes, sir. Definitely.

MG: When I initially reviewed your outline document, because this is such a massive exercise involving SHAP explainability and all these complex modules, I was honestly worried. I felt that if you tried to build all these dashboards alongside the ML models, it would take a massive amount of time for a 16-week project. But then, looking at your demo, you have already built it! So those concerns are completely resolved. I have noted these points down.

CS: Thank you, sir.

MG: And I think it is exceptionally good. You have implemented a custom loss function to enforce physics, right? That was actually one of the main questions I had prepared to ask you before your presentation. But I must say, this is a very brilliant proposal. It is highly welcome. It completely pushes the boundaries of your standard M.Tech AI projects.

CS: Thank you so much, sir.

MG: I think the best part is that you have real data. You are actively building institutional connections to get high-quality industrial data. This is excellent. You can easily publish high-impact papers from this, and you can easily scale this directly into a PhD thesis later on.

CS: Yes, sir. I will definitely do that.

MG: And because of that, I want to say all the best to you. I am looking forward to hearing about your progress. But honestly, I feel that we have already completed our Mid-Sem and In-Sem discussions. There is no need for me to sit and evaluate you separately for those rounds. I am giving you full clearance and passing marks for both your Mid-Sem and In-Sem evaluations today itself, right here.

CS: *(Amazed)* Oh... thank you so much, sir!

MG: I have never discussed a project for more than 15 to 20 minutes with any student before. This is the first time I am spending so much time discussing a proposal with someone, and that is because you are adding real, high-value contribution to the nation. People like you need to be highly appreciated. My sincere thanks to you for taking on such initiatives.

CS: Thank you, thank you so much for these kind words, sir. Hearing this from you is a very big thing for me. Thank you so much.

MG: No, no, we must appreciate and support such outstanding work. All the best to you. If you want to share your progress, please share your Git repository with me. You can keep it private if you wish, and just grant me access. That is completely fine, whichever way you prefer.

CS: Yes, sir, absolutely. I will share the repository with you. I will definitely need your guidance throughout this journey.

CS: I will share the access with you, sir. Let me know your number or contact details so I can stay in touch.

MG: Yeah, yeah. My number is **+91 9591X XXXXX**. I'll type it in the chat window. If you ever need to discuss something urgently or need guidance, you can always contact me directly on this number.

CS: 9591... Got it, sir. Thank you so much, sir.

MG: Welcome, welcome. It is genuinely my pleasure to interact with you. I am very happy to see researchers like you doing this level of work. Let's keep this discussion going, and we will move forward together. Okay?

CS: Definitely, sir. Thank you, sir.

MG: Thank you. Thanks to you as well. Have a great day. Bye-bye. And please share the presentation slides and the links to your portal with me so I can review them in detail later. Okay, bye.